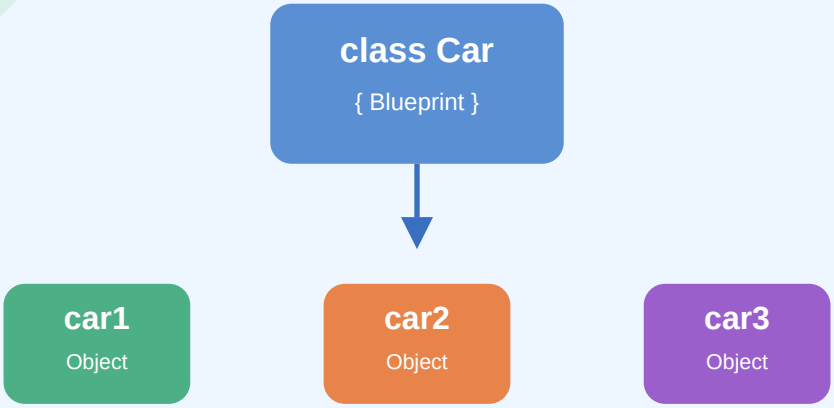
ICSE Class 9 • Computer Applications

Objects & Classes in Java

Unit 2 • Part 1



One Blueprint → Many Objects

Blueprint • Object Factory • User-Defined Types

Message Passing • Encapsulation • State & Behavior

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■ What is an Object?

An **object** is a real-world entity that has two things:
1. **State** – the data it holds (e.g., color, speed)
2. **Behavior** – the actions it can perform (e.g., start, brake)

■ Real-World Examples of Objects

Object	State (Attributes)	Behavior (Methods)
■ Car	Color, Model, Speed	start(), brake(), horn()
■ Dog	Name, Breed, Age	bark(), eat(), run()
■■■ Student	Name, Roll No, Marks	study(), write(), answer()

■ What is a Class?

A **class** is a **blueprint** or **template** for creating objects. It defines what attributes and methods an object will have. A class does not occupy memory — memory is allocated only when an object is created.

■ Class as a Blueprint – Analogy

■ Architect's Blueprint → House
Just as one blueprint can be used to build many houses, one **class** can be used to create many **objects**.

■ Biscuit Mould → Biscuits
The mould is the class. Every biscuit made from it is an object — same shape, different position!

■■ Java Syntax – Defining a Class

```
class Car {
    String color; // attribute
    String model; // attribute
    int speed; // attribute
}
```

■ Class as an Object Factory

A class works like a **factory**. You write the class **once**, and then you can create as many objects as you need using the **new** keyword.

```
Car car1 = new Car(); // Object 1
Car car2 = new Car(); // Object 2
Car car3 = new Car(); // Object 3

car1.color = "Red";
car2.color = "Blue";
car3.color = "Green";
```

■ All three objects are created from the same class **Car**, but each has its own independent data.

■ Class as a User-Defined Data Type

Primitive Data Type	User-Defined Data Type
`int x;`	`Car myCar;`
Defined by Java	Defined by YOU (programmer)
Holds simple values	Holds complex objects with data + behavior

■ Complete Java Example

```
class Dog {
    String name;
    String breed;
    int age;
}

class Main {
    public static void main(String[] args) {
        Dog d1 = new Dog();
        d1.name = "Bruno";
        d1.breed = "Labrador";
        d1.age = 3;

        Dog d2 = new Dog();
        d2.name = "Moti";
        d2.breed = "Pomeranian";
        d2.age = 2;

        System.out.println(d1.name + " is a " + d1.breed);
        System.out.println(d2.name + " is a " + d2.breed);
    }
}
```

```
Output:
Bruno is a Labrador
Moti is a Pomeranian
```

■ Three Ways to Think About a Class

■ Blueprint	■ Factory	■■ Data Type
Defines the structure of objects (attributes & methods)	Produces multiple objects — each independent, each real	A new type defined by the programmer with its own behavior

■ Quick Revision Summary

Term	Meaning
Class	Blueprint / Template / User-Defined Data Type
Object	An instance created from a class using 'new'
Attribute	Variable inside a class that stores object's state
new keyword	Allocates memory and creates an object
Dot operator (.)	Accesses attributes or methods of an object
State	Current data/values stored in an object
Behavior	Actions an object can perform (via methods)

■ Memory Trick – C.O.B.

C → **Class** is the **Cook book recipe** ■
O → **Object** is the actual **dish made** ■■
B → **Blueprint** stays same; dishes differ ■

Remember: **One Recipe** → **Many Dishes**
Just like: **One Class** → **Many Objects**

═■ Practice Questions

1. What is a class in Java? How is it different from an object?
2. Explain class as a user-defined data type with an example.

3.	Write a class **Student** with attributes: name, rollNo, marks. Create two objects and assign values.
4.	What does the **new** keyword do in Java?
5.	Why is a class called a blueprint? Give a real-life analogy.
6.	Can two objects of the same class have different data? Explain with code.
7.	Identify the class, object, attributes, and methods in: *A Bank Account has an account number, balance, and can deposit or withdraw money.*
8.	What is the dot (.) operator used for in Java?

■■ Important Tips for Exams

✔ A class does **not** occupy memory — memory is allocated only when an object is created.

✔ Each object has its own copy of the instance variables (attributes).

✔ The **new** keyword is mandatory to create an object in Java.

✔ Use the **dot (.) operator** to access attributes and methods of an object.

✔ A class can be defined anywhere in the program — order doesn't matter in Java.

✔ Class name should start with a **Capital letter** (convention: PascalCase).
